

KAIST

2026 Special Topics in Physics quantum Electronic Devices I:

Interferometers and Fractional Particles

Homework 7

Park Chanwoo

Department: Physics

Student ID: 20230297

May 23, 2026

Contents

1	Landau levels	2
(a)		2
(b)		2
(c)		3
(d)		4
(e)		4
(f)		4
2	Dirac fermions in a magnetic field	6

Landau levels

(a)

Expand the commutator using bilinearity,

$$[\Pi_x, \Pi_y] = [p_x + eA_x, p_y + eA_y] \quad (1)$$

$$= [p_x, p_y] + e[p_x, A_y] + e[A_x, p_y] + e^2[A_x, A_y] \quad (2)$$

$$(3)$$

Here $[p_x, p_y] = 0$ and $[A_x, A_y] = 0$. Also, for an analytic function F , $[p_i, F(\mathbf{x})] = -i\hbar(\partial_i F)(\mathbf{x})$. Therefore,

$$[\Pi_x, \Pi_y] = e[p_x, A_y] - e[p_y, A_x] = -i\hbar e(\partial_x A_y - \partial_y A_x) = -i\hbar eB \quad (4)$$

(b)

Introduce an operator a with real number α, β :

$$a = \alpha\Pi_x + i\beta\Pi_y \quad a^\dagger = \alpha\Pi_x - i\beta\Pi_y \quad (5)$$

the commutator is

$$[a, a^\dagger] = -2i\alpha\beta[\Pi_x, \Pi_y] = -2\alpha\beta\hbar eB \quad (6)$$

Set $\alpha = -\beta = 1/\sqrt{2\hbar eB}$ then

$$[a, a^\dagger] = 1 \quad (7)$$

Also, by definition,

$$a^\dagger a = \alpha^2\Pi_x^2 + \beta^2\Pi_y^2 + i\alpha\beta[\Pi_x, \Pi_y] = \frac{1}{2\hbar eB}(\Pi_x^2 + \Pi_y^2) - \frac{1}{2} \quad (8)$$

The cyclotron frequency of the electron is $\omega_c = eB/m^*$. Therefore,

$$H = \frac{1}{2m^*}(\Pi_x^2 + \Pi_y^2) = \frac{2\hbar eB}{2m^*} \left(a^\dagger a + \frac{1}{2} \right) = \hbar\omega_c \left(a^\dagger a + \frac{1}{2} \right) \quad (9)$$

Define a number operator $n = a^\dagger a$ (which is an Hermitian operator, $(a^\dagger a)^\dagger = a^\dagger a$). Using the commutation relation defined above, one can show that a and $a^\dagger$ satisfies the commutation relation of lowering and raising operators:

$$[n, a] = [a^\dagger a, a] = -a, \quad [n, a^\dagger] = [a^\dagger a, a^\dagger] = a^\dagger \quad (10)$$

Let W_λ be the eigenspace of n associated with eigenvalue λ . Then, for any $|\psi\rangle \in W_\lambda$,

$$na|\psi\rangle = (an - a)|\psi\rangle = (\lambda^* - 1)a|\psi\rangle \implies a|\psi\rangle \in W_{\lambda^*-1} \quad (11)$$

$$na^\dagger|\psi\rangle = (a^\dagger n + a^\dagger)|\psi\rangle = (\lambda^* + 1)a^\dagger|\psi\rangle \implies a^\dagger|\psi\rangle \in W_{\lambda^*+1} \quad (12)$$

Also, for any state $|\psi\rangle$ of the Hilbert space,

$$\langle\psi|n|\psi\rangle = \langle\psi|a^\dagger a|\psi\rangle = \|a|\psi\rangle\|^2 \geq 0 \quad (13)$$

Hence, for any nonzero eigenstate $|\psi\rangle$ of n with eigenvalue λ , there exists a non-negative integer k such that

$$a^k |\psi\rangle = 0 \quad (14)$$

If not, then we can keep applying a to $|\psi\rangle$ and get a state with negative eigenvalue, which contradicts to the fact $\langle\psi|(a^k)^\dagger n a^k |\psi\rangle \geq 0$. Let k^* be the minimal non-negative integer such that $a^{k^*} |\psi\rangle = 0$. Then, $a^{k^*-1} |\psi\rangle$ is a nonzero eigenstate of $n = a^\dagger a$ with eigenvalue $0 = \lambda - (k^* - 1)$

$$n a^{k^*-1} |\psi\rangle = a^\dagger a a^{k^*-1} |\psi\rangle = a^\dagger (a^{k^*} |\psi\rangle) = 0. \quad (15)$$

Therefore, $\lambda = k^* - 1$ is a non-negative integer.

Above proves that any eigenvalue of n is a non-negative integer. Conversely, we need to show that for any non-negative integer n can be an eigenvalue of n . To do that, assume that $|0\rangle$ is a nonzero eigenstate of n with eigenvalue 0. Problem (c) shows that such $|0\rangle$ exists. Then, for any non-negative integer N , $(a^\dagger)^N |0\rangle$ is an eigenstate of n with eigenvalue N and is nonzero:

$$\|(a^\dagger)^N |0\rangle\|^2 = \langle 0| a^N (a^\dagger)^N |0\rangle = N! \langle 0|0\rangle > 0 \quad (16)$$

For the last equality below formula is used repeatedly:

$$[a, (a^\dagger)^N] = \sum_{j=1}^N (a^\dagger)^{j-1} [a, a^\dagger] (a^\dagger)^{N-j} = N (a^\dagger)^{N-1} \quad (17)$$

$$a (a^\dagger)^N |0\rangle = (a^\dagger)^N a |0\rangle + [a, (a^\dagger)^N] |0\rangle = N (a^\dagger)^{N-1} |0\rangle \quad (18)$$

Hence, the energy eigenvalues are

$$E_n = \hbar\omega_c \left(n + \frac{1}{2} \right), \quad n = 0, 1, 2, \dots \quad (19)$$

if a ground state $|0\rangle$ exists.

(c)

The ground state $|0\rangle$ is annihilated by a ,

$$a |0\rangle = 0 \quad (20)$$

Using symmetric gauge $\mathbf{A} = \frac{B}{2}(-y, x)$, the annihilation operator is

$$a = \frac{1}{\sqrt{2\hbar eB}} \left(p_x - \frac{eB}{2} y - i p_y - i \frac{eB}{2} x \right) = -i \sqrt{\frac{\hbar}{2eB}} \left(\partial_x - i \partial_y + \frac{eB}{2\hbar} (x - iy) \right) \quad (21)$$

Introduce complex coordinate

$$z = x - iy, \quad \bar{z} = x + iy \quad (22)$$

and Wirtinger derivatives

$$\partial_z = \frac{1}{2}(\partial_x + i \partial_y), \quad \partial_{\bar{z}} = \frac{1}{2}(\partial_x - i \partial_y) \quad (23)$$

the equation $a |0\rangle = 0$ is equivalent to

$$\left(\partial_{\bar{z}} + \frac{eB}{4\hbar} z \right) \psi_g(z, \bar{z}) = 0 \quad (24)$$

Define $\phi(z, \bar{z}) = e^{\frac{eB}{4\hbar}z\bar{z}}\psi_g(z, \bar{z})$. Then,

$$\partial_{\bar{z}}\phi(z, \bar{z}) = e^{\frac{eB}{4\hbar}z\bar{z}}\left(\partial_{\bar{z}} + \frac{eB}{4\hbar}z\right)\psi_g(z, \bar{z}) = 0 \quad (25)$$

Hence, $\phi(z, \bar{z})$ is an holomorphic function of z . Therefore, the ground state wavefunction is

$$\psi_g(z, \bar{z}) = f(z)e^{-\frac{eB}{4\hbar}z\bar{z}} \quad (26)$$

where $f(z)$ is an arbitrary holomorphic function of z .

From the fact that an entire function is uniquely determined by its Taylor coefficients, the following ground state wavefunction indexed by m forms a complete basis of the ground state eigenspace:

$$\psi_{g,m}(z, \bar{z}) = N_m z^m e^{-\frac{eB}{4\hbar}z\bar{z}}, \quad m = 0, 1, 2, \dots \quad (27)$$

Where N_m is a normalization constant obtained by

$$N_m^{-2} = \int_{\mathbb{R}^2} |z|^{2m} e^{-\frac{eB}{2\hbar}|z|^2} dx dy = 2\pi \int_0^\infty r^{2m+1} e^{-\frac{eB}{2\hbar}r^2} dr = \frac{2\pi m!}{(eB/2\hbar)^{m+1}} \quad (28)$$

$$\implies N_m = \sqrt{\frac{(eB/2\hbar)^{m+1}}{2\pi m!}} \quad (29)$$

(d)

Let R_m be the root-mean-square radius of the wavefunction $\psi_{g,m}$, which is defined by

$$R_m^2 = \int_{\mathbb{R}^2} |z|^2 |\psi_{g,m}(z, \bar{z})|^2 dx dy \quad (30)$$

The magnetic flux through the disk of radius R_m is

$$\Phi_m = \pi B R_m^2 = \pi B \frac{(eB/2\hbar)^{m+1}}{2\pi m!} \int_0^\infty 2\pi r^{2m+3} e^{-\frac{eB}{2\hbar}r^2} dr = (m+1)\phi_0 \quad (31)$$

Where $\phi_0 = h/e$ is the magnetic flux quantum. Hence, there are $m+1$ magnetic flux quanta through the disk of radius R_m .

(e)

The above results show that the ground state $\psi_{g,m}$ is localized around the circle of radius R_m . For a disk sample of radius R , the state $\psi_{g,m}$ with $R_m \lesssim R$ survives while the state $\psi_{g,m}$ with $R_m > R$ is suppressed. Hence, the rough estimate of the degeneracy of the ground state is derived by the condition

$$R_{m_{\max}} \approx R \implies m_{\max} \approx \frac{\pi B R^2}{\phi_0} \quad (32)$$

Which means the number of flux quanta.

(f)

For the Landau gauge $\mathbf{A} = (0, Bx)$, the Hamiltonian is

$$H = \frac{1}{2m^*} (p_x^2 + (p_y + eBx)^2) \quad (33)$$

For the Landau gauge, the system still has translation symmetry in y -direction. Hence, we can block-diagonalize the Hamiltonian by Fourier transform in y -direction.

$$H(k_y) = \frac{1}{2m^*} (p_x^2 + (eBx + \hbar k_y)^2) \quad (34)$$

And this is the Hamiltonian of a 1D simple harmonic oscillator with the center of the potential at $x_0 = -\hbar k_y/eB$ and frequency $\omega_c = eB/m^*$. Hence, the energy eigenvalues are

$$E_n = \hbar\omega_c \left(n + \frac{1}{2} \right), \quad n = 0, 1, 2, \dots \quad (35)$$

and the ground state wavefunction up to normalization is

$$\psi_{g,k_y}(x, y) \sim e^{ik_y y} e^{-\frac{(x-x_0)^2}{2l_B^2}} \quad (36)$$

where $l_B = \sqrt{\hbar/eB}$.

the density of states along k_y -direction is $d_y/2\pi$ where d_y is the length of the sample in y -direction. Also, the finite sample size in x -direction restricts the range of k_y by $-eBd_x/2\hbar \lesssim k_y \lesssim eBd_x/2\hbar$. Hence, the degeneracy of the ground state is approximately

$$\frac{d_y}{2\pi} \cdot \frac{eBd_x}{\hbar} = \frac{Bd_x d_y}{\phi_0} \quad (37)$$

This also means that the number of flux quanta through the sample is the degeneracy of the ground state.

Dirac fermions in a magnetic field

Start from the Hamiltonian of a 2D Dirac fermion in a magnetic field,

$$H = v(\Pi_x \sigma_x + \Pi_y \sigma_y) \quad (38)$$

Introduce same ladder operators as in problem 1,

$$a = \frac{1}{\sqrt{2\hbar qB}} (\Pi_x \pm i\Pi_y), \quad a^\dagger = \frac{1}{\sqrt{2\hbar qB}} (\Pi_x \mp i\Pi_y) \quad (39)$$

The sign in the definition of a and $a^\dagger$ depends on the sign of qB . For example, if $qB < 0$ the sign is chosen as same as in problem 1, while if $qB > 0$ the sign is chosen as opposite to that in problem 1.

With properly chosen sign, the ladder operators satisfy the commutation relation $[a, a^\dagger] = 1$ and the argument in problem 1 can be reused.

Then, the Hamiltonian can be rewritten as

$$H = v\sqrt{2\hbar qB}(a^\dagger \sigma_+ + a \sigma_-) \quad (40)$$

Where $\sigma_\pm = (\sigma_x \pm i\sigma_y)/2$ is the raising/lowering operator of the spin degree of freedom.

Choose a normalized ground state $|0\rangle$ annihilated by a and define $|n\rangle$ by

$$|n\rangle = \frac{(a^\dagger)^n}{\sqrt{n!}} |0\rangle \quad (41)$$

Then, $|n\rangle$ is normalized (Equation. 16) eigenstate of $a^\dagger a$ and can be raised/lowered by $a^\dagger$ and a :

$$a |n\rangle = \frac{1}{\sqrt{n!}} a (a^\dagger)^n |0\rangle \quad (42)$$

$$= \frac{1}{\sqrt{n!}} \left((a^\dagger)^n a + [a, (a^\dagger)^n] \right) |0\rangle \quad (43)$$

$$= \sqrt{n} |n-1\rangle \quad (\because \text{Equation 17}) \quad (44)$$

$$a^\dagger |n\rangle = \frac{1}{\sqrt{n!}} a^\dagger (a^\dagger)^n |0\rangle \quad (45)$$

$$= \frac{1}{\sqrt{n!}} (a^\dagger)^{n+1} |0\rangle = \sqrt{n+1} |n+1\rangle \quad (46)$$

For $n \geq 1$, construct the following state

$$|n, \pm\rangle = \frac{1}{\sqrt{2}} (|n\rangle \otimes |\uparrow\rangle \pm |n-1\rangle \otimes |\downarrow\rangle) \quad (47)$$

Then, $|n, \pm\rangle$ is an eigenstate of H with eigenvalue $\pm v\sqrt{2\hbar qBn}$:

$$H |n, \pm\rangle = v\sqrt{2\hbar qB} (a^\dagger \sigma_+ + a \sigma_-) \frac{1}{\sqrt{2}} (|n-1\rangle \otimes |\uparrow\rangle \pm |n\rangle \otimes |\downarrow\rangle) \quad (48)$$

$$= \frac{v\sqrt{2\hbar qB}}{\sqrt{2}} (a^\dagger |n-1\rangle \otimes |\downarrow\rangle + a |n\rangle \otimes |\uparrow\rangle) = \pm v\sqrt{2\hbar qBn} |n, \pm\rangle \quad (49)$$

Also, the state $|0\rangle \otimes |\uparrow\rangle$ is an eigenstate of H with eigenvalue 0:

$$H |0\rangle \otimes |\uparrow\rangle = v\sqrt{2\hbar qB} a |0\rangle \otimes |\downarrow\rangle = 0 \quad (50)$$

Hence, the energy spectrum includes $\pm v\sqrt{2\hbar q B n}$ for non-negative integer n .

Considering H^2 ,

$$H^2 = 2\hbar q B v^2 \begin{pmatrix} a^\dagger a & 0 \\ 0 & a a^\dagger \end{pmatrix} = 2\hbar q B v^2 \begin{pmatrix} a^\dagger a & 0 \\ 0 & a^\dagger a + 1 \end{pmatrix} \quad (51)$$

The spectrum of H^2 is $2\hbar q B v^2 n$ for non-negative integer n . This implies that $\pm v\sqrt{2\hbar q B n}$ are the only possible eigenvalues of H . Hence, the spectrum of H is $\pm v\sqrt{2\hbar q B n}$ for non-negative integer n .